Shreyas Govinda Raju

Backend, Data & ML Platform Engineer

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Summary

Software engineer with **8+ years at Amazon** building distributed backend services, ML infrastructure, and Amazon's public open-source developer SDKs. Leads cross-team platform initiatives, owning architecture from design through production, root-causing performance regressions that logs and dashboards cannot explain, and mentoring engineers. Cut ML serving cost by **90%** with a GPU migration.

Technical Skills

Languages Python, Java, TypeScript, JavaScript, SQL, Bash

Distributed Systems Microservices, event-driven architecture, asynchronous processing, API design, concurrency, rate limiting, observability

Cloud & Infrastructure AWS (ECS, Fargate, Lambda, SQS, EventBridge, Step Functions, S3, DynamoDB, IAM), Docker, AWS CDK, CI/CD

ML & Data PyTorch, GPU model serving, Apache Spark / PySpark, XGBoost, Amazon Redshift, AWS Glue, OpenSearch

APIs & Tools GraphQL, Swagger / OpenAPI code generation, Git

Experience

Amazon

Seattle, Washington

Software Development Engineer 2 - Machine Learning & Data Platform — *May 2024 - Present*

- Led the architecture and delivery of a production **ML inference platform** for two product teams with a shared public API, request-state store, distributed work queue, and auto-scaling serving fleet, launched in one quarter by a team of 4 and sustaining peak loads of **50 TPS**.
- Architected high-throughput data pipelines to productionize multiple batch ML models, processing **1 billion records daily** to serve online and offline predictions.
- Reduced inference p90 latency from 5-15 s to **200 ms (>95%)**, replaced an auto-scaled fleet of 30-60 CPU nodes with **6-10 GPU tasks**, and cut serving compute cost by **90%** by building the business case for a CPU-to-GPU migration and securing finance sign-off before any spend.
- Drove engineering standards across the organization: led **cross-team architecture and design reviews**, established the platform's observability and on-call practices, and mentored engineers and interns, including a data engineer who built the weekly **300M US / 1B EU** embedding index.
- Root-caused a fleet-wide latency regression that logs and dashboards could not explain: replayed production payloads and used a control model on identical hosts to isolate three causes, including processor-generation differences (**AVX-512 vs AVX2**).
- Scaled an ML scoring platform from **1 to 5 regions** across 10+ international marketplaces, owning orchestration, feature-generation, and **PyTorch** model-training pipelines supporting new-product discovery.
- Reduced new-marketplace onboarding from weeks to a configuration change by decomposing a monolithic training pipeline into modular, **configuration-driven workflows** with per-marketplace and per-model overrides.
- Led standardization of **Spark** job packaging and deployment across 8 production pipelines, aligning owners across teams and writing the org-wide migration guide.

Software Development Engineer 2 - Commerce Platform (Buy with Prime) & Partner Integrations — *Jun 2022 - May 2024*

- Led the architecture of an **event-integration framework** for Buy with Prime partners (event subscriptions, rate-limited queues, pluggable handlers, output adapters), cutting partner onboarding from about **3 months to under a week**. About 10 merchants run on it, at roughly **100 events/s combined (8M+ a day)**.
- Built the framework's **data-enrichment layer**, a typed API client over Order, Inventory and Availability **GraphQL** queries with secure credential management, converting sparse event notifications into complete business objects.
- Engineered delivery guarantees and burst protection: per-order **FIFO ordering**, idempotent consumers, bounded retries, 14-day dead-letter redrive, and **concurrency controls** across the queue-to-compute path, with queue-depth and dead-letter alarms for integration owners.
- Proposed the SDK as a scoped, reversible step while webhooks were blocked on other teams, and aligned domain teams and solutions architects on the plan. Partner feedback then shaped the webhook design.
- Automated partner-facing **API documentation** from the live schema, replacing a manual process and keeping internal-only fields out.

Software Development Engineer - Alexa Skills Kit, Developer Tools & Public SDKs — *Jul 2018 - Jun 2022*

- Core contributor to Amazon's open-source **Alexa Skills Kit SDKs** for Python, Node.js and Java and the ASK CLI, published to **PyPI, npm and Maven** and used by third-party developers.
- Owned the **API specification and code-generation pipeline** for three language SDKs, where a single API change propagated across every published client library, and enforced release boundaries to prevent unreleased functionality from reaching public SDKs.
- Designed a **declarative dialog framework** for the Node.js SDK, replacing state machines with a composable control tree for initiative, confirmation, and validation.
- Built the public **Skill Management API SDK for Python** from scratch, including authentication, response modeling, testing, and packaging.
- Shipped core **ASK CLI** capabilities across two major versions, including project scaffolding, build hooks, API commands, and local debugging.

Education

University of Southern California, Los Angeles

Aug 2016 - May 2018

M.S. in Computer Science

National Institute of Technology Karnataka, Surathkal, India

Jul 2011 - May 2015

B.Tech in Computer Science and Engineering